

Joshua Yorko

SITE RELIABILITY ENGINEER · SOFTWARE ENGINEER

Worcester, MA

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“Site Reliability Engineer and Kubernetes Specialist dedicated to enhancing system resilience and operational excellence.”

Skills

Programming Languages	Python (Expert), Bash, SQL, JavaScript, TypeScript
Infrastructure & DevOps	Terraform, AWS, GCP, CI/CD (GitHub Actions), Rancher, GitOps
Data Engineering	Python, Apache Iceberg, DuckDB, Spark, Dremio, Nessie,
API Development	FastAPI, Django API Integration & Management, Microservices Architecture
Virtualization & Containerization	EC2, Harvester HCI (SUSE), Proxmox, Docker, Kubernetes, K3s, RKE2, EKS
GitOps	ArgoCD, Flux, Fleet
Full Stack Development	Django, HTML/CSS, JavaScript, React, Next.js, PostgreSQL, Tailwind, Pocketbase
Artificial Intelligence	OpenAI API, AWS BEDROCK, Anthropic, Langchain, Ollama, Llama3

Experience

Gainwell Technologies, Conway, AR

SOFTWARE ENGINEER/SRE

June 2023 - Present

- Designed and executed CI/CD pipelines using GitHub Actions, streamlining deployment processes for enhanced efficiency.
- Orchestrated containerized application deployment via Kubernetes on AWS EKS, optimizing for auto-scaling and high availability.
- Developed RESTful APIs with FastAPI and managed their EKS deployment via Terraform, establishing a resilient container environment.
- Automated text extraction in AWS S3 using Amazon Textract and AWS Lambda, centralizing data into a data lake for improved access and analysis.
- Managed AWS infrastructure using Terraform, overseeing services such as EC2, S3, RDS, Elastic Load Balancers, and VPCs.
- Integrated Large Language Models via Amazon Bedrock into data analytics processes for enhanced natural language understanding capabilities.
- Built a microservice images API on FastAPI and Kubernetes to transform deprecated Dental DataClearingHouse APIs into a usable API for frontend UI, delivering clear JSON responses via a custom CDN with PostgreSQL and S3 object store.
- Built an open data lakehouse on EKS with Dremio and Nessie, transforming raw data from an IBM3270 mainframe into an Apache Iceberg data lake and feeding a Python-based work distribution algorithm for dental adjudicators.

Gainwell Technologies, Conway, AR

SENIOR DATA ENGINEER/SOFTWARE ENGINEER

June 2022 - March 2023

- Architected an enterprise-level data lake utilizing AWS Glue, Amazon S3, PySpark, Hadoop, Python, and PostgreSQL, significantly boosting data processing efficiency.
- Integrated the data lake with Amazon RDS via SQL views for a Django frontend, enabling seamless data access and manipulation.
- Automated complex data retrieval using Python libraries (Requests, aiohttp, asyncio, Selenium, BeautifulSoup), enhancing data acquisition speed and reliability.
- Developed real-time dashboards in PowerBI and custom visualizations using Plotly and Django, delivering insightful data analytics to stakeholders.
- Ensured high availability and scalability of web applications hosted on Amazon EC2.
- Implemented robust security measures, including SSL encryption and firewalls, to safeguard data and applications.
- Proactively monitored applications and infrastructure, minimizing downtime and maximizing operational reliability.

Lambent Spaces, Boston, MA

IMPLEMENTATIONS SPECIALIST

March 2021 – October 2022

- Directed software implementations, ensuring customer requirements were met through effective network and system configurations.
- Optimized deployment processes, resulting in heightened efficiency and improved customer satisfaction.
- Fostered collaboration with cross-functional teams to refine deployment workflows and procedures.
- Authored and utilized SQL and Python scripts for automation and efficient problem resolution.

CVS HEALTH, Cumberland, RI

DATA ANALYST (DESKTOP SUPPORT)

May 2019 – March 2021

- Designed and deployed O365 Apps for Enterprise across 80k devices, leveraging Python and Excel for efficient reporting and achieving a 30% increase in deployment efficiency.
- Created and maintained business-focused reports using Dash/Plotly, SharePoint, and Excel, driving informed decision-making processes.
- Conducted extensive data normalization from SCCM and Active Directory, ensuring data accuracy and consistency.
- Played a pivotal role in Windows 10 Enterprise Migration, focusing on rigorous testing and quality assurance of developed reports.

SPENCER TECHNOLOGIES, Medway, MA

CLIENT ONBOARDING SPECIALIST

Jan 2017 – Apr 2019

- Implemented the Smartsheet Request Management System, resulting in annual cost savings of \$16k.
- Successfully onboarded new business ventures worth over \$15 million, enhancing company growth and market presence.
- Developed and executed project plans for multiple IT projects, leading teams to successful and timely client launches.

Education

Indiana State University

Terre Haute, IN

B.A. IN ENGLISH

Aug 2006 – June 2008

RPA & Automation Projects

ROBOCORP MIGRATION & ORCHESTRATION

2023 – Present

- Migrated legacy UiPath robots to Python using open-source Robocorp tools and RCC, reducing licensing costs and increasing maintainability.
- Maintained a personal fork of RCC (v18.5) with custom enhancements, extending functionality and contributing back to the open-source community.
- Designed high-throughput producer/consumer robots leveraging prebuilt RCC holotree environments, reducing robot startup times by 75% and enabling efficient distributed processing.
- Developed a custom orchestration platform using **yorko-io/control-room** to manage and monitor robot execution across Kubernetes clusters.
- Developed high-performance Playwright-based web-browsing robots with optimized startup performance, significantly reducing end-to-end processing times.

Cloud & Containerized Robotics

KUBERNETES & DEVCONTAINERS

2023 – Present

- Built custom **arc-runner** Docker images with prebuilt holotree environments to enable rapid scale-out of robotics workloads on Kubernetes, improving deployment speed by 60%.
- Leveraged devcontainers (**room-of-requirement**) to create reproducible development and execution environments across local and cloud infrastructure.
- Integrated Docker with RCC Remote for seamless remote execution and debugging of Robocorp robots.
- Implemented scalable architecture using GitHub Actions self-hosted ARC runner sets, eliminating the need for Redis queues by adopting file adapters and file sharding.
- Designed matrix job sharding strategies to distribute workloads efficiently across multiple GitHub runners.

Open Source Contributions

AUTOMATION TOOLS & PLATFORMS

2023 – Present

- Contributed to open-source projects including **fetch-repos-bot**, **sema4ai-action-server**, and other automation tools, enhancing functionality and reliability.
- Maintained several GitHub repositories demonstrating advanced automation systems, such as **control-room**, **arc-runner**, and related orchestration components.
- Actively developed **control-room**, a Robocorp-inspired orchestration platform designed to manage robot lifecycles and workload distribution.